Week 03 – Lab Assignment

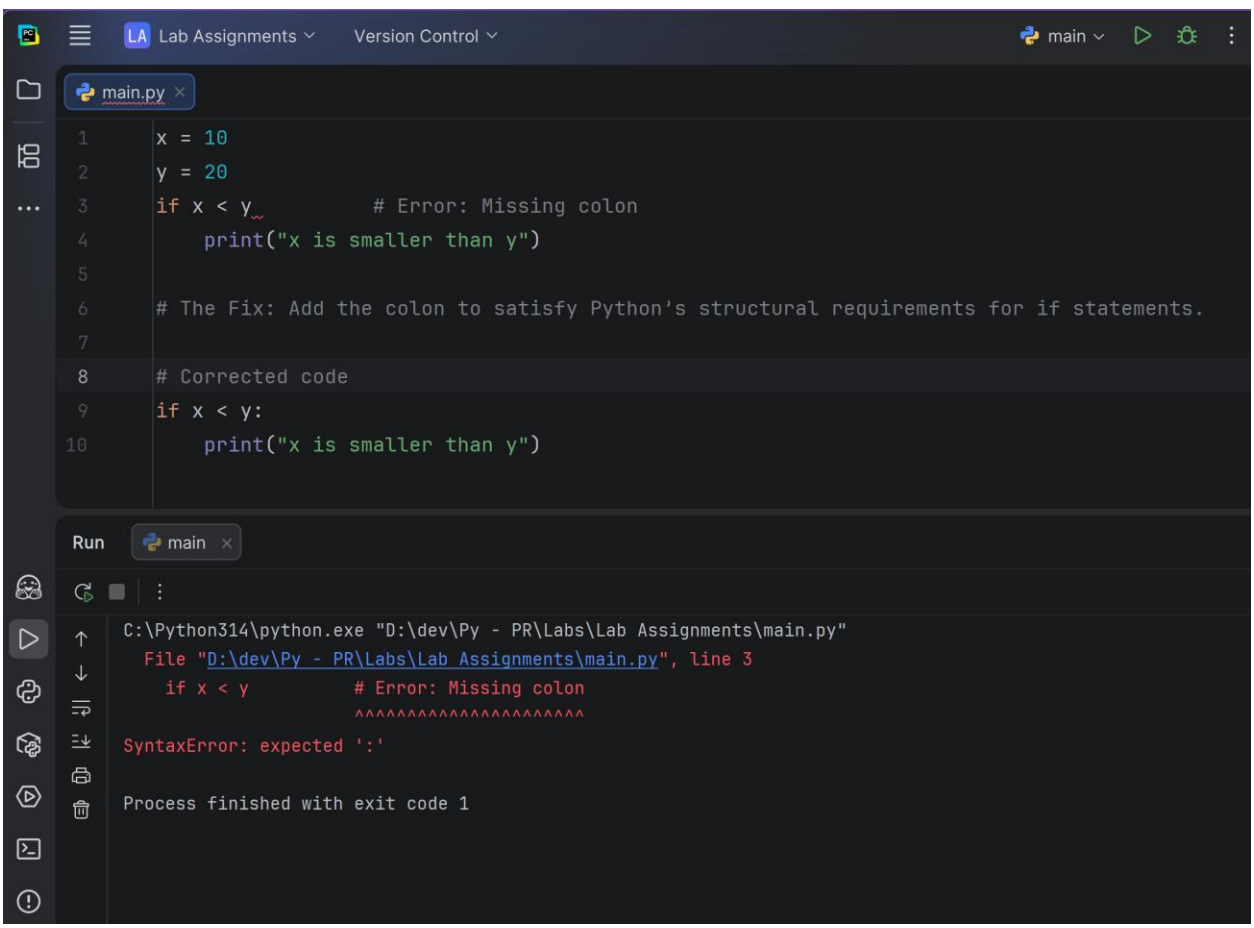
Write Python scripts that include intentional syntax, runtime, and semantic errors, and practice debugging them.
Below are three short Python scripts, each one intentionally contains one type of error.

1. Syntax Error

Definition: A Syntax Error occurs when the code violates the grammatical rules of the Python language. The interpreter cannot parse the script, so the program fails to start.

Intentional Error: Missing a colon (:) at the end of a control statement.

Code Example:

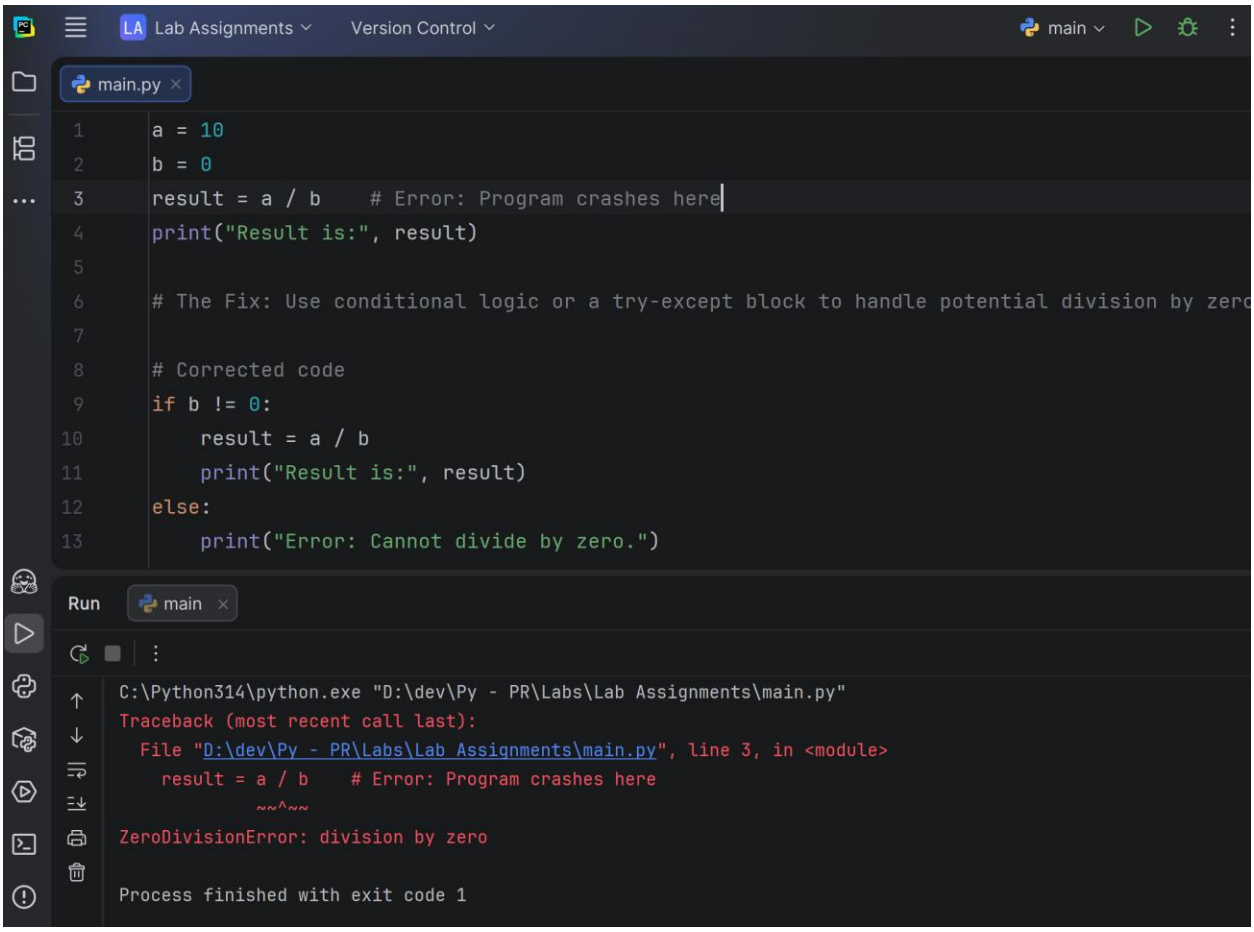


2. Runtime Error Example

Definition: A Runtime Error occurs while the program is running. The syntax is correct, but an operation is attempted that the computer cannot execute (e.g., dividing by zero or referencing a non-existent file).

Intentional Error: Attempting to divide an integer by zero (ZeroDivisionError).

Code Example:

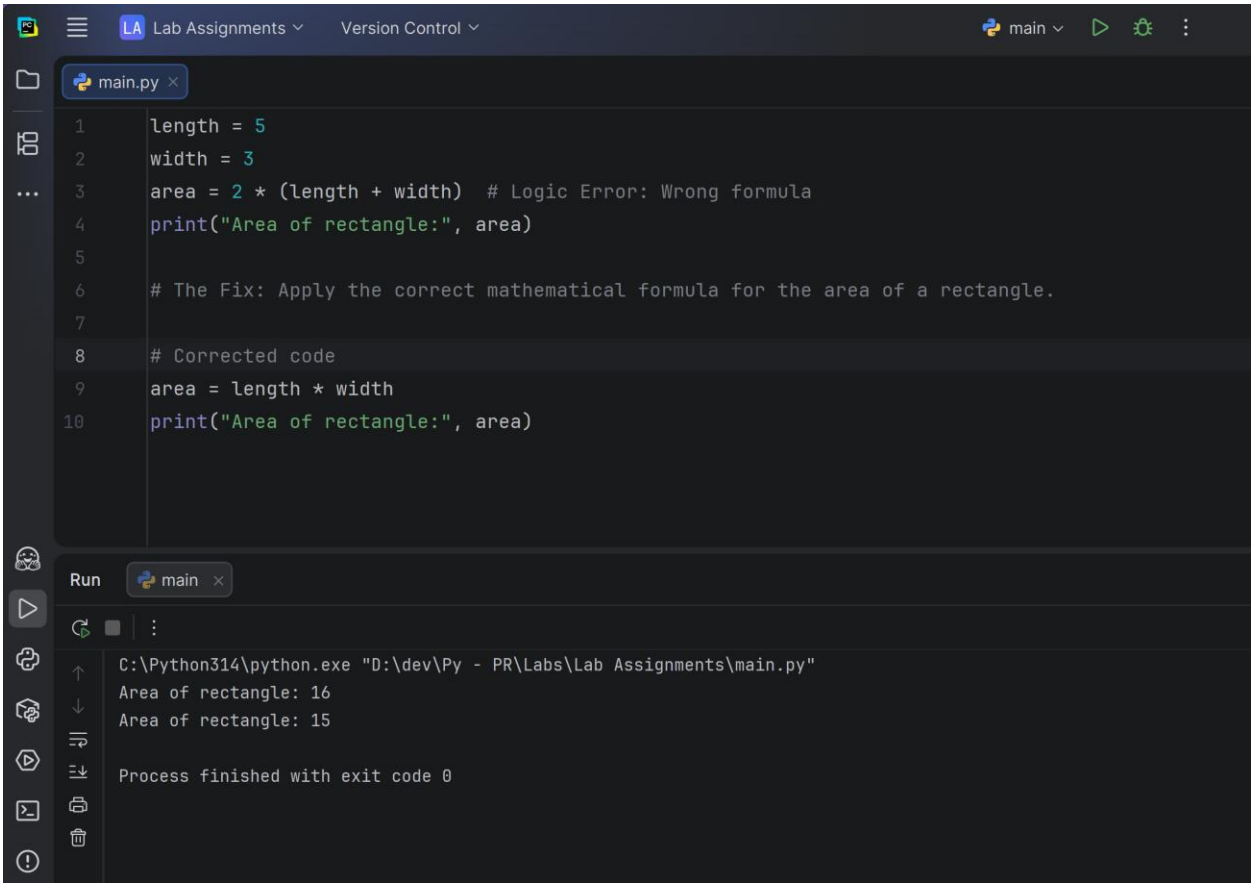


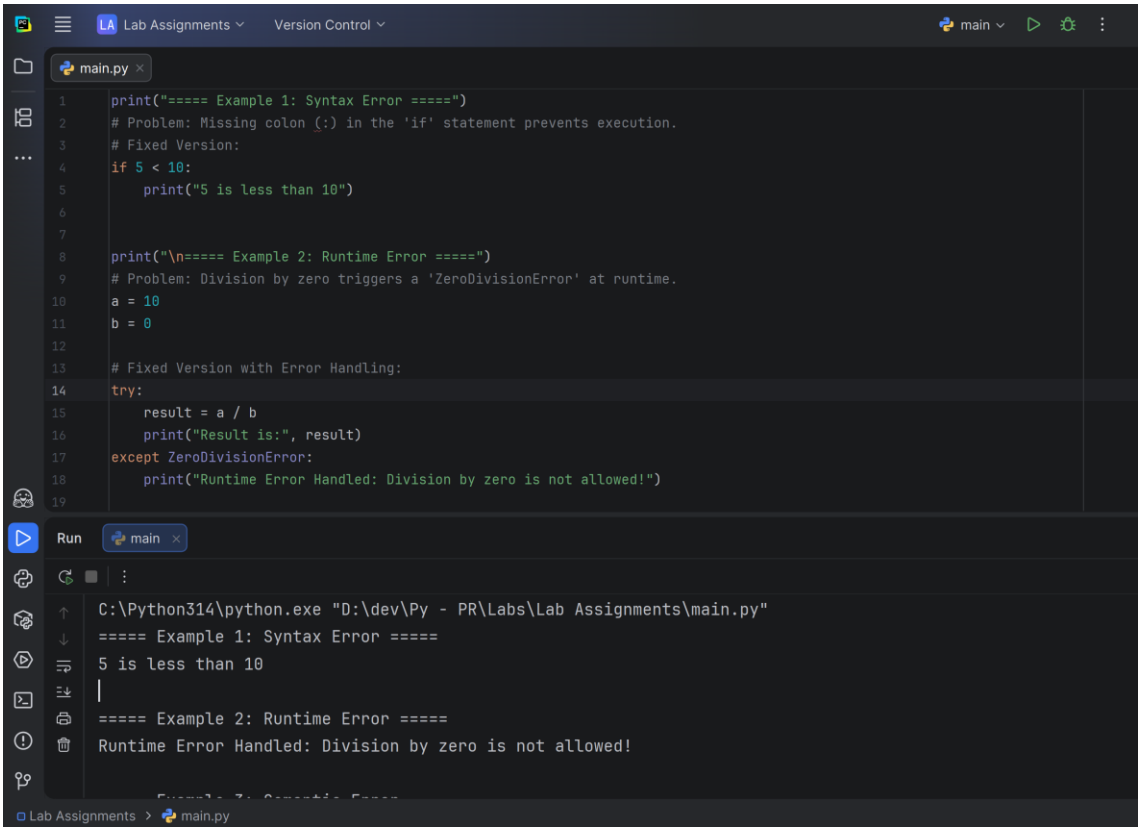
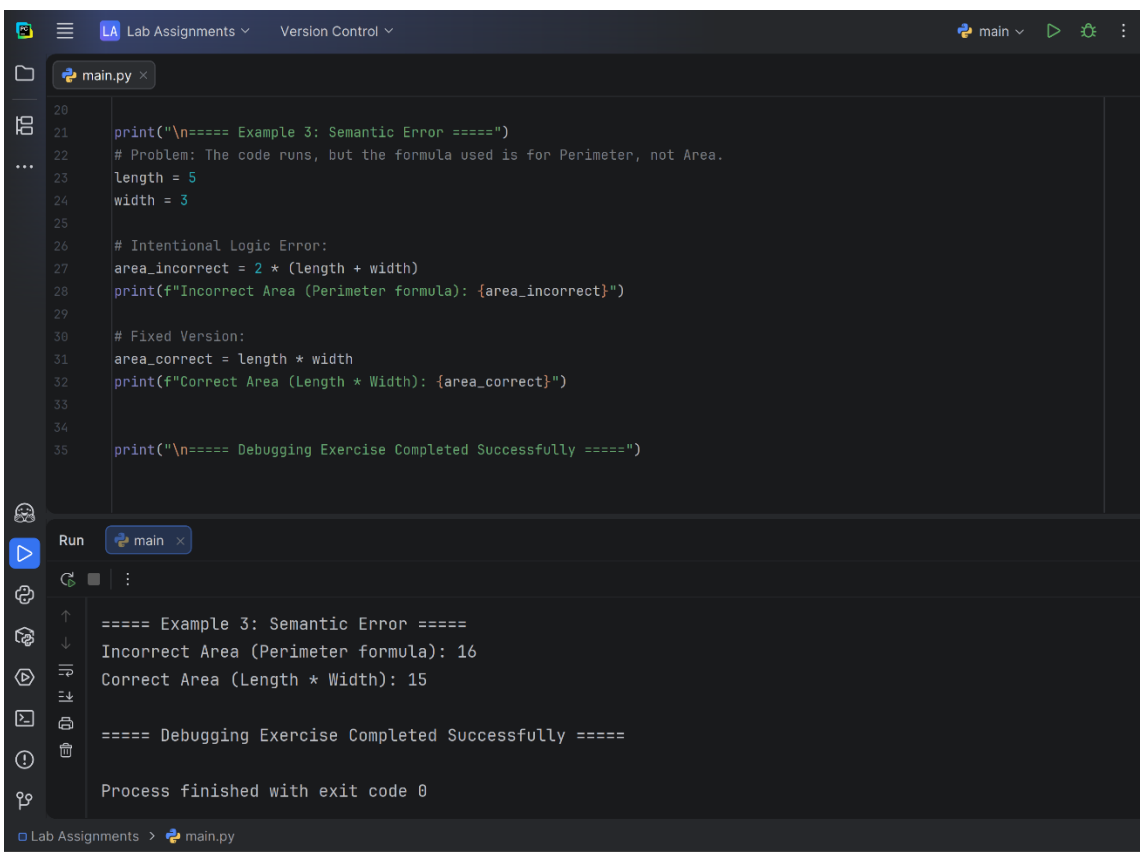
3. Semantic Error Example

Definition: A Semantic Error (or Logic Error) occurs when the code runs without crashing but produces an incorrect or unintended result due to flawed logic in the algorithm.

Intentional Error: Using the perimeter formula when the goal is to calculate the area.

Code Example:



Complete Debugging Script

Below is the unified script incorporating all examples with their respective fixes and error handling.